

BRAC UNIVERSITY
Department of Computer Science and Engineering
Summer 2025
CSE 421: Computer Networks
Assignment 1
Marks : 5

CO1	Name:	ID:	Section:
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Question 1

[1]

In a fully connected mesh topology, there are 8 devices interconnected with each other. Calculate the total number of links required in this network.

Question 2

[1]

Draw a hybrid topology with these topologies: 1 bus, 2 stars and 2 ring topology. [You can use n no. of devices]

Question3

[1+1]

Alice and Bob are both downloading a large file. Alice is using a **torrent application**, where she downloads parts of the file from multiple users and also shares parts she already has with others. Bob, on the other hand, is streaming a movie from **Netflix**, where a central server provides the content directly to his device.

1. Identify which system follows the **Peer-to-Peer (P2P)** model and which follows the **Client-Server** model.
2. Explain how both models work and compare their advantages and disadvantages.

Question 4

[1]

List two differences between Port address and IP address. What are they called together?